

TCG Tracker

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TCG Tracker

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Problem

Most TCG apps support only one game, forcing users to switch apps and split collections. Different apps have different features.

Gap: No simple mobile app supports multiple TCGs in one unified system.

Solution

Scan, track, and manage all cards in one unified collection app.

Similar Apps

Pokémon TCG Live (single-game only), Decked Builder (complex MTG tools), Collectr (tracking-focused).

TCG Tracker combines scanning + collection + decks across multiple games.

Challenges & Solutions

API inconsistency → Unified card data into a consistent multi-game schema.

Premium API limitations → Added local mock cards mirroring real API structure.

Database complexity → Designed a scalable Room DB structure for multiple TCGs.

Learnings

- Simpler UI design improves usability more than adding features
- Consistent data models are critical when merging multiple APIs

Future

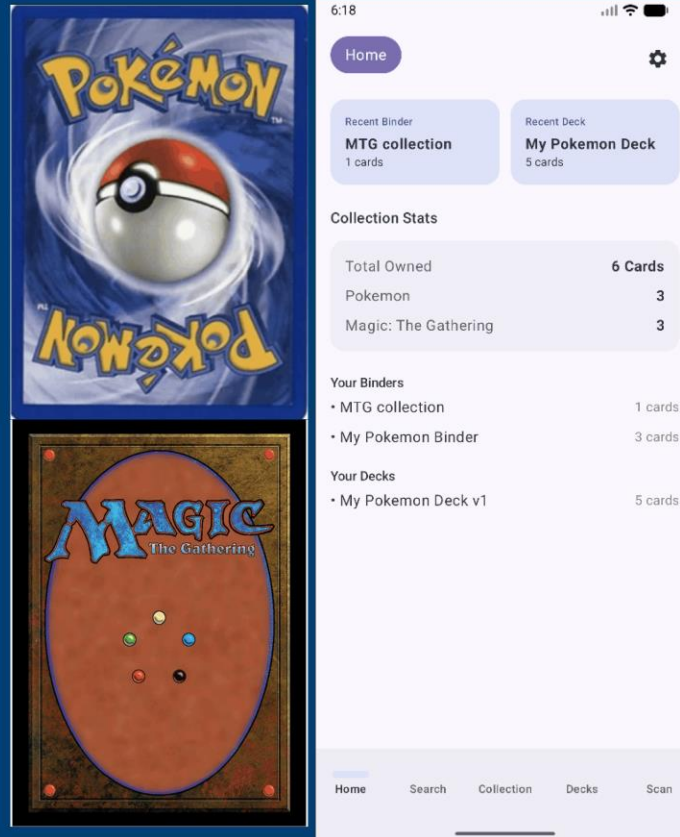
More TCG game support and collection exporting.

One app. Every card.

Scan. Search. Organize. Build decks.

TCG Tracker is a mobile app for scanning and managing trading card collections across multiple games in one place.

For: Pokémon and MTG collectors who want a single unified system.



Smart Search

Card Name

Pokémon MTG

Shared Filters

Set / Expansion

Card Number

Rarity

Any Common Uncommon Rare

Mythic

Pokémon Filters

Type

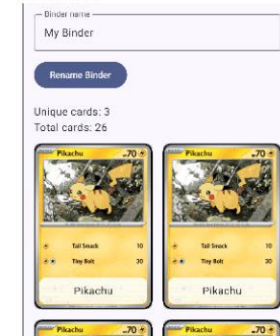
Any Grass Fire Water

Lightning Psychic Fighting Darkness

Scan



Collection



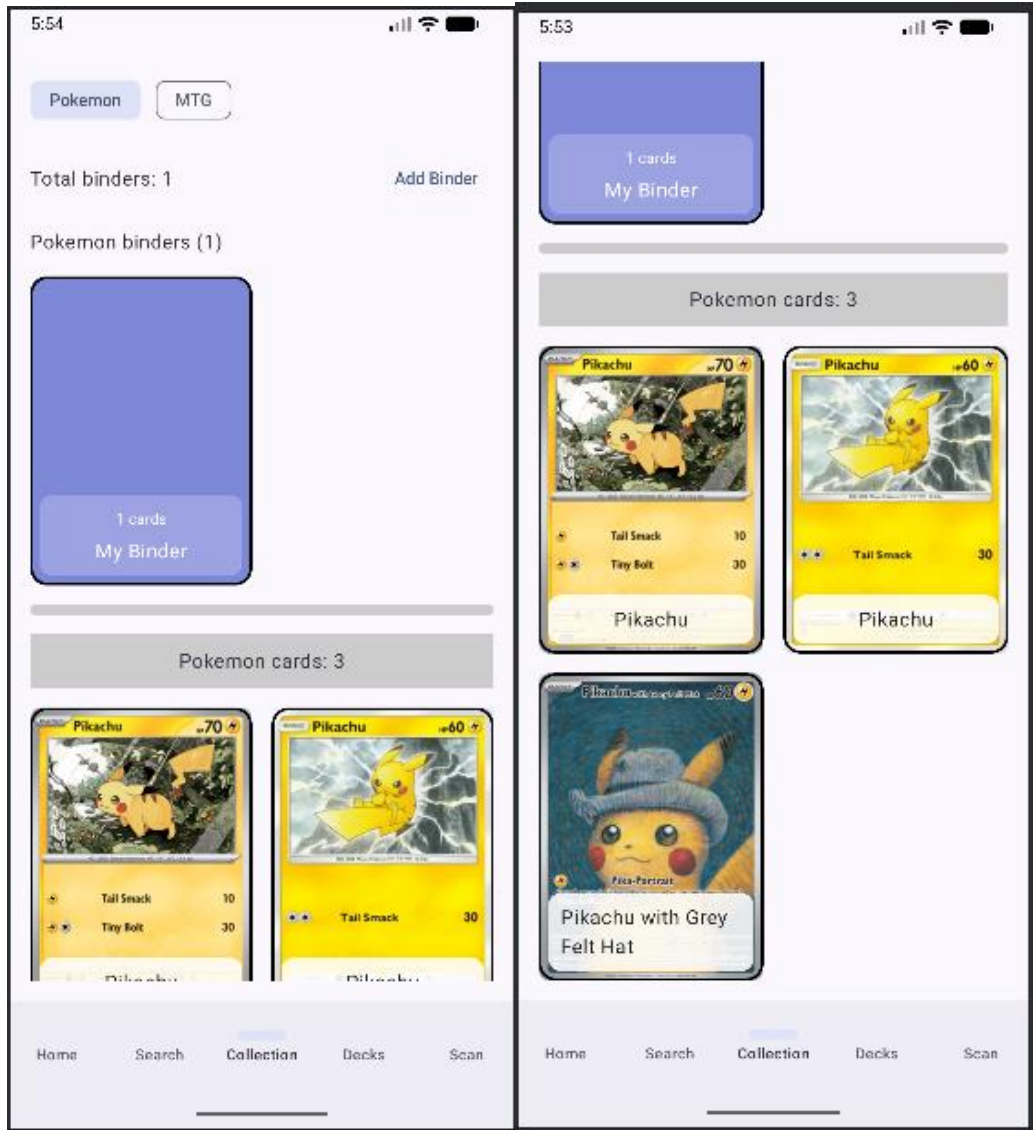
The problem

- Trading Card Games (TCG) have thousands of unique cards available, so manual tracking is extremely messy.
- Michael is an active TCG player, and cannot find a single software that helps put all of this info into one place. And apps that do are very messy
- Most existing tracker apps are specialized to one specific TCG

Motivation and goals

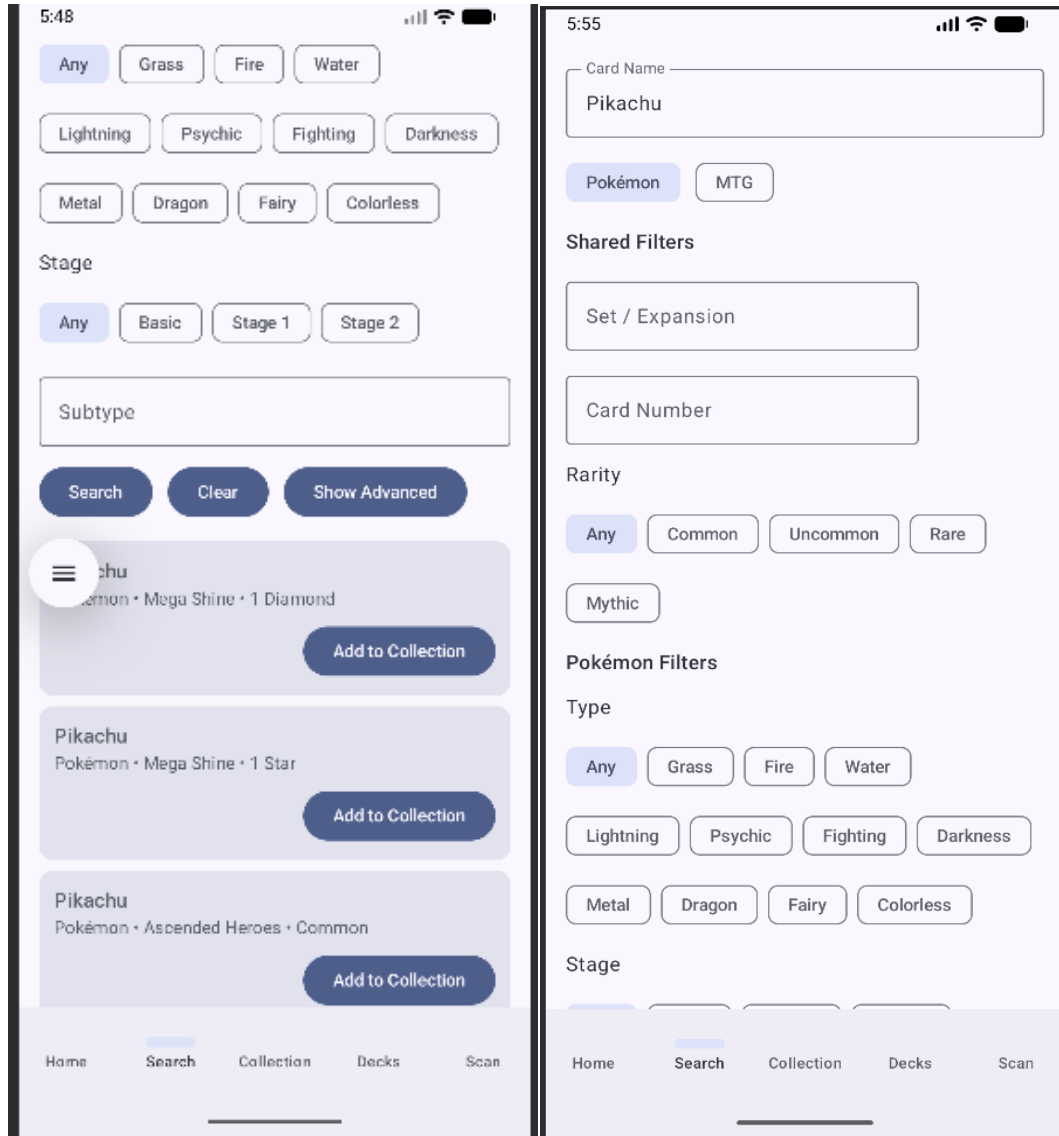
- We wanted to create a centralized and efficient app that is built to support multiple TCGs at once, and allow the user to easily switch back and forth
- We wanted to start with support for a few TCGs at first, and build it in a way that is easily expandable
- Michael had already designed the UI/UX mockups for a very similar project in a previous class

Our App



- The user can adjust the quantity of the cards added, and can create binders and decks within the TCG.
- The decks enforce standard playing rules of the specific TCG
- For example, a Pokemon deck must have exactly 60 cards, and no more than 4 duplicates of each card.

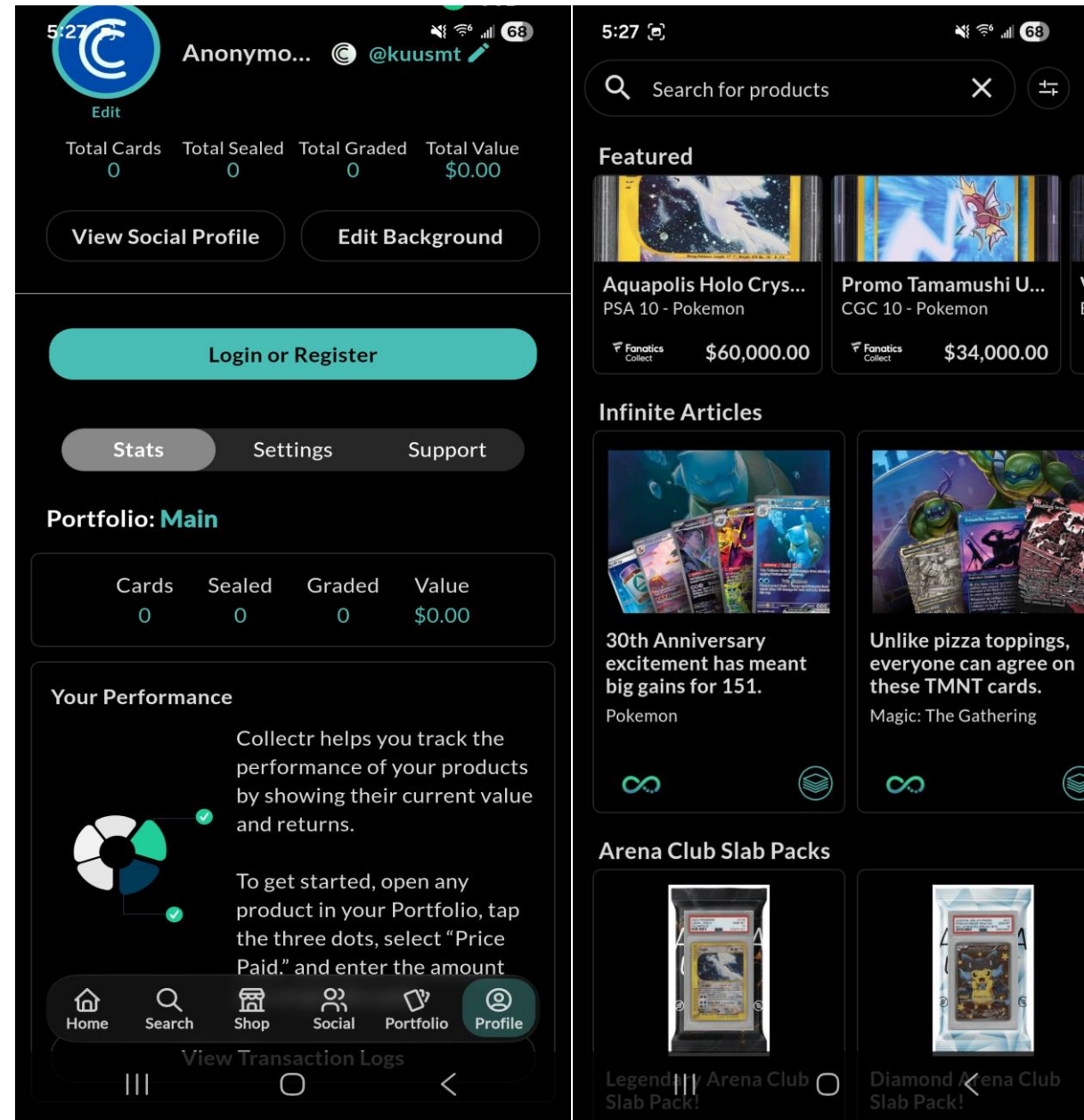
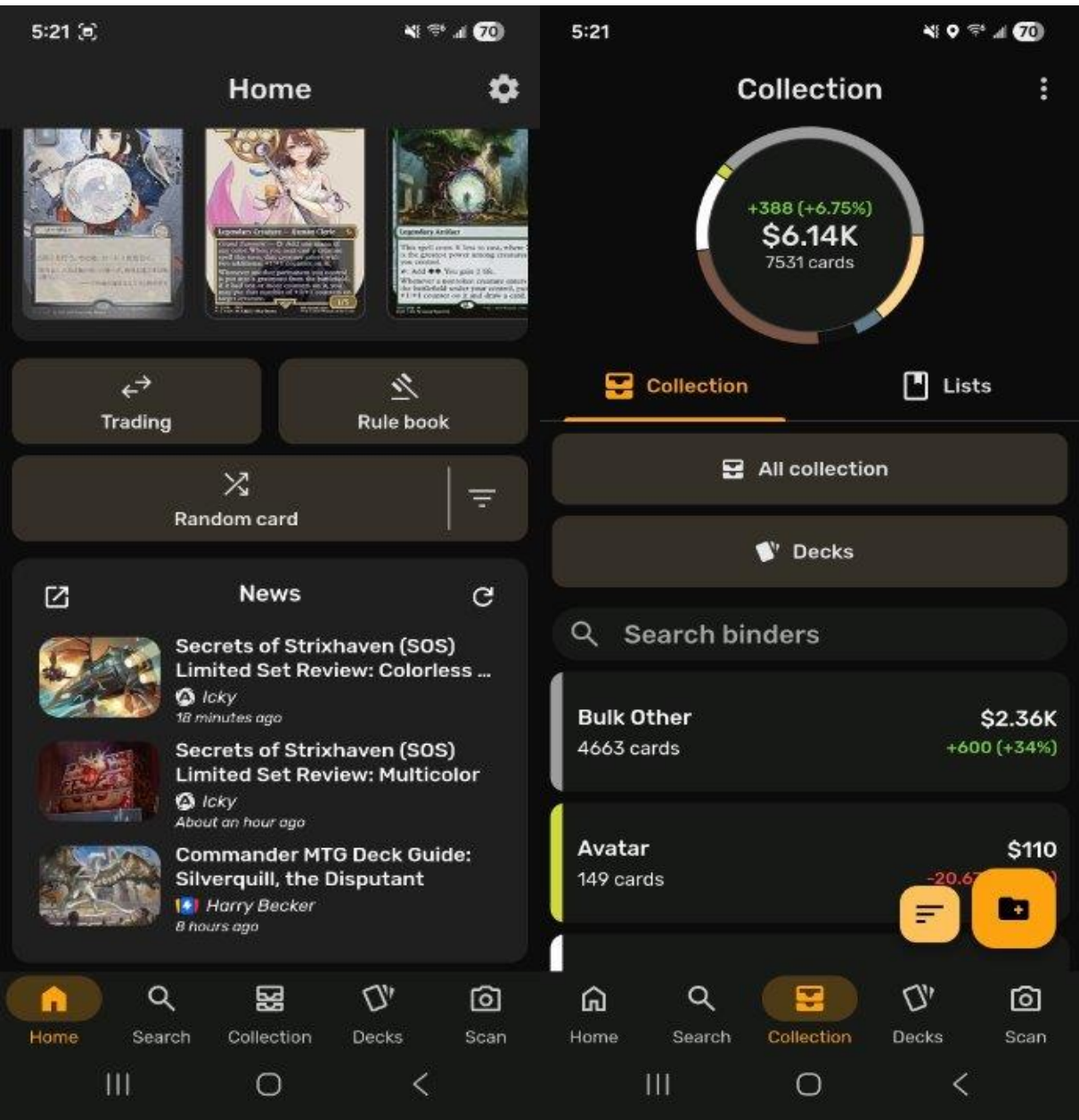
Our App (cont)



- Our app allows users to search or scan cards, and add them to their collection
- Search screen with advanced filters
- Camera scan will automatically pull the card name from the photo and let the user add the card to their collection

Similar apps on the market

Feature	CardHub	Collectr	Manabox
Multi-Game	Yes	Yes	No
Price Tracking	No	Yes	Yes
Deck Making	Yes	No	Yes
Selling	No	Yes	No



Challenges and solutions

- Getting advanced search filters to map to the API properly. Eachgame has **very** different search parameters. Solution: make separate UI layouts for each TCG
- Getting the camera scan to work well with our API
- We used a premium API, so testing was difficult. Solution: add temporary "mock" data that mirrored the API, and only gets used if the API access token is not setup.

What we learned

- **Our biggest failure:** We originally wanted the database and API calls to be dynamic, so that any TCG can be used. But we learned very quick how different the API calls and card responses (JSON formatted) are for each TCG.
- For this project, we decided to use 2 TCGs (Magic the Gathering and Pokemon), with differnt API call builders and search screens for each.
- **Technology learned:** API calls, DAOs, database storage/access, complex screen navigation, themes (light/dark mode), ViewModels, lazy lists, top and bottom app bars, camera usage, and more.

What could be expanded

- More TCG options
- Decks could factor in banned cards
- Decks could have better rule editing
- Card value (\$) tracking

Demo Video

- https://youtu.be/MP2DXR_wyKk

Questions?